

Lab Exercise: Schematics with ESP32

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PART A – Circuit Calculations

You are required to calculate:

1. Total resistance
2. Total current from the power supply
3. Current through each resistor
4. Voltage drop across each resistor
5. Power dissipated through each resistor

Useful formulas:

Ohm's Law

$$V = IR$$

Where:

- V = Voltage (Volts)
- I = Current (Amps)
- R = Resistance (Ohms)

Power Formula

$$P = VI$$

Circuit A (Series Circuit)

The circuit contains:

- 20Ω resistor
- 30Ω resistor
- 50Ω resistor
- 12V power supply

Since the resistors are connected in series:

$$R_T = R_1 + R_2 + R_3$$

1. Calculate Total Resistance

$$R_T = 20 + 30 + 50$$

$$R_T = 100$$

2. Calculate Total Current

Using Ohm's Law:

$$I = V/R$$

$$I = 12/100$$

$$I = 0.12A$$

In a series circuit, current is the same everywhere.

3. Current through each Resistor

- Current through $20\Omega = 0.12A$
 - Current through $30\Omega = 0.12A$
 - Current through $50\Omega = 0.12A$
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4. Voltage Drop Across Each Resistor

For 20Ω resistor

$$V = IR$$

$$V = 0.12 * 20$$

$$V = 2.4V$$

For 30Ω resistor

$$V = 0.12 * 30$$

$$V = 3.6V$$

For 50Ω resistor

$$V = 0.12 * 50$$

$$V = 6V$$

$$2.4 + 3.6 + 6 = 12V$$

5. Power Dissipated

20Ω resistor

$$P = VI$$

$$P = 2.4 * 0.12$$

$$P = 0.288W$$

30Ω resistor

$$P = 3.6 * 0.12$$

$$P = 0.432W$$

50Ω resistor

$$P = 6 * 0.12$$

$$P = 0.72W$$

Final Answers – Circuit A

Quantity	Answer
Total Resistance	100Ω
Total Current	$0.12A$
Current through 20Ω	$0.12A$
Current through 30Ω	$0.12A$
Current through 50Ω	$0.12A$
Voltage across 20Ω	$2.4V$
Voltage across 30Ω	$3.6V$

Quantity	Answer
Voltage across 50Ω	6V
Power in 20Ω	0.288W
Power in 30Ω	0.432W
Power in 50Ω	0.72W

Circuit B (Parallel Circuit)

The circuit contains:

- 20Ω resistor
- 100Ω resistor
- 50Ω resistor
- 12V power supply

Because this is a parallel circuit:

$$1/R_T = 1/R_1 + 1/R_2 + 1/R_3$$

1. Calculate Total Resistance

$$1/R_T = 1/R_1 + 1/R_2 + 1/R_3$$

$$1/R_T = 0.05 + 0.01 + 0.02$$

$$1/R_T = 0.08$$

$$R_T = 1/0.08$$

$$R_T = 12.5 \Omega$$

2. Calculate Total Current

$$I = V/R$$

$$I = 12/12.5$$

$$I = 0.96A$$

3. Current Through Each Resistor

In parallel circuits, voltage across each resistor is the same.

Therefore each resistor gets 12V.

Current through 20Ω resistor

$$I = 12/20$$

$$I = 0.6A$$

Current through 100Ω resistor

$$I = 12/100$$

$$I = 0.12\text{A}$$

Current through 50Ω resistor

$$I = 12/50$$

$$I = 0.24\text{A}$$

$$0.6 + 0.12 + 0.24 = 0.96\text{A}$$

4. Voltage Across Each Resistor

In a parallel circuit:

- Voltage across $20\Omega = 12\text{V}$
 - Voltage across $100\Omega = 12\text{V}$
 - Voltage across $50\Omega = 12\text{V}$
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5. Power Dissipated

20Ω resistor

$$P = VI$$

$$P = 12 * 0.6$$

$$P = 7.2\text{W}$$

100Ω resistor

$$P = 12 * 0.12$$

$$P = 1.44\text{W}$$

50Ω resistor

$$P = 12 * 0.24$$

$$P = 2.88\text{W}$$

Final Answers – Circuit B

Quantity	Answer
Total Resistance	12.5Ω
Total Current	0.96A

Quantity	Answer
Current through 20Ω	0.6A
Current through 100Ω	0.12A
Current through 50Ω	0.24A
Voltage across 20Ω	12V
Voltage across 100Ω	12V
Voltage across 50Ω	12V
Power in 20Ω	7.2W
Power in 100Ω	1.44W
Power in 50Ω	2.88W
